

NANYANG TECHNOLOGICAL UNIVERSITY**SEMESTER 1 EXAMINATION 2022-2023****IE4476 – IMAGE PROCESSING**

November / December 2022

Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 3 pages.
2. Answer all 4 questions.
3. All questions carry equal marks.
4. This is a Restricted Open Book Examination. You are allowed to bring into the examination hall one double-sided A4-size reference sheet with texts handwritten or typed on the A4 paper (no sticky notes/post-it notes on the reference sheet).
5. Unless specifically stated, all symbols have their usual meanings.

1. The response $g(x, y)$, obtained when a filter is applied to a digital image $f(x, y)$, is specified by

$$g(x, y) = 4f(x, y) - f(x - 1, y) - f(x + 1, y) - f(x, y - 1) - f(x, y + 1).$$

- (a) Find the impulse response $h(x, y)$ of the filter and list two spatial properties of the filter.

(6 Marks)

- (b) Find the frequency response $H(u, v)$ of the filter and state a frequency property of the filter.

(7 Marks)

- (c) The filter is applied to a 9×9 digital image $f(x, y)$ as shown in Figure 1 on Page 2. Show the 9×9 output image with the assumption that $f(x, y) = 0$ outside Figure 1.

Note: Question No. 1 continues on page 2.

0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	0	0
0	0	1	1	1	1	1	0	0
0	0	1	1	1	1	1	0	0
0	0	1	1	1	1	1	0	0
0	0	1	1	1	1	1	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0

Figure 1

(7 Marks)

- (d) From the result in part (c), discuss the filtering effects.

(5 Marks)

2. A 5×5 image $f(x, y)$ is given in Figure 2.

1	0.75	0.75	0.75	1
0.75	0.5	0.25	0.5	0.75
0.75	0.25	0	0.25	0.75
0.75	0.5	0.25	0.5	0.75
1	0.75	0.75	0.75	1

Figure 2

- (a) Design a coding scheme to code the image with the minimum number of bits.

(8 Marks)

- (b) Compute the average number of bits per pixel for your designed coding scheme in part (a). What is the number of bits per pixel required if all the pixels are coded using uniform-length coding?

(5 Marks)

- (c) Show the image $g(x, y)$, which is enhanced from $f(x, y)$ using histogram equalization and is kept within the range of 0.04 to 1.

(7 Marks)

- (d) Find the histograms of $f(x, y)$ and $g(x, y)$.

(5 Marks)

3. A binary image is represented by set A .
- Design an algorithm to extract an 8-connected component from a binary image represented by set A .
(7 Marks)
 - Design an algorithm to extract a 4-connected component from a binary image represented by set A .
(4 Marks)
 - Let set A contain the shaded pixels in Figure 3. The pixel indicated by “0” in Figure 3 is selected to extract an 8-connected component. Write the value n in the proper pixel cells/locations to indicate the sequence of which the pixels are extracted in the n^{th} iteration of your algorithm.

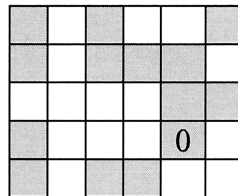


Figure 3

- (7 Marks)
- Show the result for part (c) for extracting a 4-connected component instead of 8-connected component.
(7 Marks)
4. An edge detector has detected 4 pixels from image $f(x, y)$ at x - and y -coordinates of $(-1, 1)$, $(0, 2)$, $(1, 1)$, and $(2, 0)$. A Hough transform model $y = ax + b$ is applied to detect possible lines, where a and b are continuous variables.
- Find all the possible values of the resulting Hough transform $H(a, b)$.
(5 Marks)
 - Find the positions of the largest and the second largest values of $H(a, b)$ in the ab -plane.
(8 Marks)
 - Plot the Hough transform with the horizontal axis a and the vertical axis b .
(7 Marks)
 - Find the conditions for a and b such that $H(a, b) > 0$.
(5 Marks)

END OF PAPER

IE4476 IMAGE PROCESSING

Please read the following instructions carefully:

- 1. Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**
2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.
3. Please write your Matriculation Number on the front of the answer book.
4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.